

Fascia and musculoskeletal injury: An underestimated association?

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Introduction

The prevention and treatment of musculoskeletal disorders represent superordinate goals of training in sports. Extensive research efforts have therefore been made to identify and describe the most frequent injuries occurring during athletic activity. However, while numerous studies meticulously report the incidence and prevalence of muscle, bone, tendon, ligament or capsular injuries, the deep fascia remains a Cinderella tissue in this regard: it is largely unknown to what extent it can be damaged although its vulnerability is highly plausible. This chapter gives two examples of the involvement of fascia in the genesis and symptomatology of musculoskeletal disorders, highlighting its potential role in soft tissue strain injury and delayed onset muscle soreness.

Soft tissue strain injuries

Muscle injuries are one of the most frequent health problems in sports. According to epidemiological data, 10–55% of all acute injuries sustained during athletic activity are muscle strains (Chan et al., 2012). The pathology is particularly prevalent in sports such as football (soccer), basketball, rugby and running (Maffulli et al., 2015). Its typical location is in the lower limb as the quadriceps femoris, gastrocnemius, as well as the adductor and hamstring muscles (the latter particularly in football), represent the most commonly damaged structures (Barroso and Thiele,

2011; Ekstrand et al., 2011). The susceptibility of the listed muscles can be ascribed to a variety of features: among others, a high percentage of glycolytic fast twitch muscle fibers, pennate fiber arrangement and bi-articularity have been suggested to be predisposing factors (Maffulli et al., 2015).

Pathophysiology of soft tissue strains

Powerful decelerations requiring eccentric contraction (see Chapter 26) represent the main pathomechanism of injury (Maffulli et al., 2015). This implies that the affected tissue has to counteract high elongating forces in the presence of active muscle work. While a certain amount of stress is tolerated by a resilient muscle-tendon complex, its load-bearing capacity can be exceeded by repetitive and cumulative stimuli, finally leading to rupture. In particular, end game situations in sports represent a dangerous period for athletes. In football, the number of injuries rises almost proportionally until the end of the two halves (Ekstrand et al., 2011), which seems to be explained by accumulating exhaustion. Results from animal experiments revealed that the force necessary to provoke muscle failure is reduced by ~3–7% in the fatigued state (Mair et al., 1996). This, in turn, is of relevance because biomechanical studies in human athletes demonstrated that eccentric strength, probably representing a crucial protective factor, decreases with increasing playing time (Grieg and Siegler, 2009).

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The role of fascia in soft tissue strains

Although clinical diagnostics and associated therapies traditionally focus on the skeletal muscle and the tendon, it is not the only actuator that may be strained during injury. The collagenous connective tissue forms a functional and structural unit with the active component of the locomotor system. This can be observed on both a micro and a macro level. The micro level includes the connectivity of the individual fibers with the endomysium as well as another surprising architectural feature: in some muscles, as few as one fifth of the fibers extend over the entire distance between origin and insertion, whereas the rest of the fibers terminate within the muscle belly, being linked through their endomysium (Figure 16.1; Hijikata and Ishikawa, 1997). Hence, the connective tissue appears to represent a relevant mechanical hub transmitting and distributing forces within the muscle and can be stretched during eccentric contraction. The macro level, in addition to the micro-arrangement, includes the tight linkages of muscles with their deep fascia as well as the tissue fusion with adjacent (e.g. synergists and antagonists) and serially connected (e.g. gastrocnemius and hamstrings) structures (for a more detailed summary on myofascial continuity, see Chapter 12). Taken together, the intimate micro- as well as macro-level fusions of the connective tissue with the skeletal muscles suggest

that the former assists the latter mechanically by helping to absorb and damp peak stresses produced during explosive movements such as accelerations, stops, or changes of direction. The hypothesis that the deep fascia may be injured in athletic movements, thus appears tenable and merits further contemplation.

The findings of a recent systematic review with meta-analysis (Wilke et al., 2019) corroborate the assumed vulnerability of fascial tissues during sports. Analyzing a total of 1433 muscle strain injuries registered in 15 observation studies (diagnosis verified by magnet resonance imaging and/or ultrasound), it found 9 out of 10 lesions included structural damage to the fascia or the tendon. While the deep fascia exhibited tears in 32% of the cases, an even higher share was detected for the myotendinous junction (68%), which seems highly plausible in view of the force transmitting function of the tendon. In contrast to these surprisingly high values, isolated damage to the muscle occurred only in 1 out of 10 injuries (11%). The term muscle injury, therefore, ignores the morphological substrate of the disorder and may better be replaced by “myofascial injury” or “myocollagenous injury”.

Fascia-specific therapy of muscle injuries and new approaches for prevention

From a theoretical point of view, both sports performance and the duration of the rehabilitation

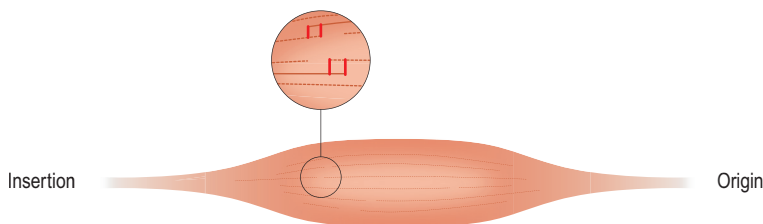


Figure 16.1 Schematic illustration of a skeletal muscle

While spanning muscle fibers bridge the entire distance between insertion and origin, non-spanning fibers (circle) are linked only by means of connective tissue (red vertical lines).

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process may be adversely affected by connective tissue damage detected in soft tissue injuries. In fact, after fasciotomy (a longitudinal incision of the fascia), Garfin et al. (1986) found force decreases of up to 16% in the leg muscles of dogs. The authors suggested that the force loss was related to substantial (~50%) reductions in intra-compartmental pressure. A recent study made similar observations: removal of the deep fascia surrounding the wing muscles of wild turkeys decreased the maximal force output by 28% compared with the pre-surgery state (Ruttiman et al., 2018). It therefore remains to be elucidated if the typical size of fascial ruptures in myocollagenous injury is sufficiently large to have similar consequences in athletes. In addition to force production, fascial lesions could also affect joint range of motion. Bishop et al. (2018) examined the effect of a 2 cm-long injury on tissue thickness and shear strain mobility of the porcine lumbar fascia. The incision led to a slightly higher thickness and a decrease of ~15% in fascial layer sliding. Both factors may impair range of motion (Wilke et al., 2018). Interestingly, the magnitude of the changes was markedly more pronounced if the animals were subjected to an 8-week immobilization period. Although the downtimes following soft tissue strains are not as long, the reciprocal aggravation of injury and lack of movement could also be of relevance in humans.

With regard to the potential impact of fascial injury on the duration of the rehabilitation process, the available data is hitherto insufficient. Some observation studies registered a higher return to play time in case of greater fascial damage (Prakash et al., 2018; Werner et al., 2017). Other studies (Connell et al., 2004; Ekstrand et al., 2016; Pedret et al., 2015; Pollock et al., 2016) compared return to play after myofascial and myotendinous injuries. Mostly, they revealed similar

durations for both types, which seems expected as both are characterized by collagen failure. Although not distinguishing tendinous and fascial lesions, one study investigated the potential difference between myocollagenous and purely muscular soft tissue strain injuries. In a sample of 70 elite athletes, Renoux et al. (2018) demonstrated that myocollagenous lesions required a 3-week longer rehabilitation process than isolated muscle lesions (7.6 vs. 3.9 weeks). While this finding needs to be solidified by further studies, it is congruent with the fact that the connective tissue has a substantially higher turnover rate than the skeletal muscle (Heinemeier et al., 2013).

Despite the lack of data clearly delineating the impact of fascial tissue on return to play time, a focused treatment providing specific stimuli to the connective tissue could still speed up recovery when assuming it had not been optimal to date. Basically, two factors determine the resilience of the connective tissue to strain forces: the relative amount of type I collagen and the arrangement of the fibers. To stimulate the production of type I collagen and to elicit an optimal fiber orientation (in tendons parallel, lattice-like in fascia), the application of mechanical forces lengthening the targeted tissue is required. As an example, this can be achieved by means of eccentric training and stretching. Eccentric exercise is already widely used in current rehabilitation paradigms. Both acutely and in the long term, it leads to a substantial shift of the optimal force production angle towards longer muscle lengths (Brockett et al., 2001). The enhanced mechanical output in stretched positions may represent a protective mechanism against strain injury. When designing preventive interventions, clinicians should particularly consider the eccentric-concentric strength ratio. Croisier et al. (2002) examined this outcome in a sample of athletes with injury history and recurrent strains.

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Most of the study participants had substantial bilateral deficits in eccentric strength. After a focused eccentric training program, the imbalance was restored and none of the athletes suffered from a recurrence during the observation period of 12 months. In general, three meta-analyses have examined the effectiveness of eccentric training in the prevention of hamstring strains. All conformably conclude that the exercises, if performed with high compliance, have a prophylactic impact (Al Attar et al., 2017; Goode et al., 2015; Schache, 2012).

With regard to the effectiveness of stretching, there is a paucity of evidence. Findings from fundamental research basically support its application, suggesting a crucial role in regulating the inflammatory response and tissue remodeling. Corey et al. (2012) stretched previously inflamed lumbar mice fasciae twice daily for 10 minutes. After 12 days, they observed a normalization of the step length, less macrophage expression, and a reduced mechanical sensitivity. In a follow-up trial by the same research group, a similar treatment protocol was applied to examine the acute impact of stretching (Berrueta et al., 2016). Forty-eight hours after induction of connective tissue inflammation, rats which had performed flexibility exercises displayed a decreased thickness of the inflammatory lesion and had a smaller neutrophil count. While increased collagen production following the inflammatory phase is an essential part of functional healing, an overshooting synthesis can lead to fibrosis and excessive scarring. Interestingly, a 20–30% subcutaneous connective tissue stretch in mice has been shown to decrease the injury-induced expression of the transforming growth factor beta 1 (TGF- β 1) and procollagen type-1. The production of both substances can lead to fibrosis and, with this, to adhesions reducing tissue functionality and flexibility. It has to be

reiterated that a large elongation, like the one applied in the experiment, may be difficult to achieve in humans under *in vivo* conditions: according to biomechanical testing, the normal deep fascia ruptures at strains of 12–15% (Wang et al., 2009; Yahia et al., 1993). However, with a continuous lengthening protocol, progressively performed over several consecutive days, the failure threshold can increase by up to 27–37% (Wang et al., 2009). Another issue relates to the optimal timing of when to apply a stretching intervention within the rehabilitation period as it cannot be extrapolated from the experimental data.

Despite its potentially beneficial impact on tissue regeneration, stretching should be used cautiously. If the exercise stimulus is too strong, the positive effect may vanish or even reverse. Comparing two stretch intensities, Wang et al. (2007) found that rabbit leg lengthening at a rate of 2 mm per day caused collagen fibril necrosis whereas 1 mm per day induced regenerative changes. Furthermore, the higher stretching rate was shown to foster collagen type III production (Wang et al., 2008). This collagen type is typically synthesized after trauma and provides provisional, but less stable, tissue healing. Under normal circumstances, the deep fascia consists almost exclusively of type I collagen, which is directly related to the tensile strength of a tissue. The shift towards a higher type III share may therefore be the result of an excessive strain rate.

Only one study has investigated the value of an explicit stretching exercise program on the success of the return-to-play process. Malliaropoulos et al. (2004) examined the recovery of 80 elite athletes with hamstring strain injuries. The study participants were randomly allocated to two groups: while the first group

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stretched once a day, the second group performed four stretching sessions. The intensified protocol led to a 2-day faster restoration of the normal range of motion and a significantly shorter return-to-play process (13.3 vs. 15.1 days) compared with the controls, who were stretching less. Although more trials are needed to solidify the evidence for the effectiveness of stretching as a treatment of soft tissue strain injuries, it may at least be used as an adjunct element of a multifaceted rehabilitation program.

Similar to its application with therapeutic purposes, stretching is also often used as a prophylactic measure. Several systematic reviews clearly showed that it does not reduce the overall occurrence of injuries (Thacker et al., 2004; McHugh & Cosgrave, 2010; for a complete review see Chapter 10). However, there appears to be a small beneficial effect if only muscle injuries are considered (McHugh & Cosgrave, 2010). More importantly, the available studies mostly used static stretching, which may not provide sufficient repeated lengthening stimuli.

In summary, specific cyclic stretching interventions with higher volumes tailored to mainly load the extramuscular fascia (see Chapter 10) could open new frontiers in the prevention of myocollagenous injuries. Myocollagenous healing may also be improved using nutritional supplementation. Orally ingested vitamin C-enriched gelatin and rope-skipping exercises have been shown to enhance collagen synthesis *in vivo* (Shaw et al., 2016). Interestingly, there seems to be a linear dose response relationship because the intake of 15 g elicited a stronger response than the use of 5 g. Although no effects on mechanical stability were found, it was further demonstrated in a placebo-controlled trial that supplementation with collagen peptides (5 g) can improve self-perceived functional stability of the

foot in participants with chronic ankle instability (Dressler et al., 2018).

Delayed onset muscle soreness

Delayed onset muscle soreness (DOMS) is a well-known phenomenon for both couch potatoes as well as high-level athletes getting back to action after the off-season: when training loads are rapidly increased or new, previously unfamiliar exercise stimuli are applied, motion-related muscle pain can develop on the days after activity. It has long been thought that DOMS primarily originates from the muscle. In fact, compelling evidence suggests that eccentric exercise in particular results in increased levels of creatine kinase, an enzymatic marker of muscle damage. Findings from biopsy studies, furthermore, revealed structural changes such as damage of z-disks and disruption of myofibers (Stauber et al., 1990). Taken together, these processes are suggested to lead to tissue inflammation, swelling and pain, representing the characteristic hallmarks of DOMS (Cleak and Eston, 1992; Kanda et al., 2013).

Despite the proven existence of muscle damage following severe or unaccustomed exercise, recent research has challenged the classical theory about the pathogenesis of DOMS, showing that the delayed discomfort felt after exercise does not seem to stem from the muscle. Gibson et al. (2009) required their participants to perform eccentric contractions of the tibialis anterior muscle. One day later, hypertonic saline (a solution eliciting pain) was injected into the muscle as well as into the surrounding fascia. Interestingly, the pain ratings of the participants were substantially higher upon fascial injection, while the nociceptive response to muscle irritation was not even different to the non-exercised control leg. A similar study by Lau

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et al. (2015) confirmed these surprising findings. After inducing DOMS in the elbow flexors, the pain reaction of different tissues was assessed by means of the electrical pain threshold, which identifies the current intensity corresponding to the first pain sensation. Again, the stimulation of the surrounding fascia was demonstrated to be more painful than the irritation of the muscle. From the data of Gibson et al. (2009) and Lau et al. (2015), it can be inferred that the deep fascia (and not the muscle itself) appears to represent the source of nociception in DOMS. This hypothesis fits with the data of Barry et al. (2015), who found the lumbar fascia to exhibit a 3-fold higher nerve density than the back muscles.

With regard to the pathophysiology of DOMS, the evidence pointing towards the existence of structural damage in the skeletal muscle and pain perception stemming from the fascia enables two slightly different explanation models (Langevin & Sherman, 2006; Wilke et al., 2017). First, when assuming structural muscle damage as a cause, the nociceptors of the connective tissue surrounding it may be stimulated by the inflammation-induced muscular swelling. Connected to this, it is also conceivable that the lesion of the muscle would trigger an increased sensitivity of the fascia due to its innervation by the same spinal segment. Then even light stimuli such as touch or moderate muscle contractions could evoke pain. Another theory completely negates the pathogenic value of the muscle damage, which may then rather be considered an essential part of the remodeling/adaptation process following training. In this case, both origin and manifestation of pain could be in the connective tissue. As the fascia is intimately connected to the skeletal muscle and needs to adapt to its volume and length changes, it may be damaged particularly during heavy-load eccentric contractions (see above). To date, the presence

of micro-ruptures within the deep fascia has not been investigated explicitly. However, after 50 maximal eccentric contractions leading to DOMS, increased excretion of urinary hydroxyproline and hydroxylysine have been reported (Brown et al., 1997). Both amino acids are suggested to reflect collagen breakdown and, with this, possibly also lesions of the intra- or extramuscular connective tissue. In another study (Crameri et al., 2004), biopsies were taken from the human vastus lateralis muscle. Unaccustomed high-intensity eccentric exercise led to tenascin C staining and the detection of reactive macrophages in the perimysium and epimysium. Tenascin C is a surrogate of high loading and/or failure of a tissue, whereas the presence of macrophages is indicative of inflammatory processes. Finally, in a recent experiment, ultrasound imaging was used to quantify the thickness of the biceps brachii fascia in response to fatiguing eccentric exercise (Nosaka & Lau, 2018). On the day after the induction of DOMS, thickness was described to be ~30% higher. During subsequent days, further increases were registered, reaching the maximum (+104%) on the fifth day. Although there are multiple potential causes for the observed fascial thickening, it may be the result of injury-provoked edema. Taken together, the experimental findings appear to reveal that exercise causing DOMS does not only provoke structural muscle damage but also micro-lesions within the fascia. It is, hence, tenable to assume that the mechanisms underlying the two hypotheses presented above act in concert when muscle soreness occurs following physical activity.

Summary

Musculoskeletal traumas can substantially affect the deep fascia of the skeletal muscle. This becomes highly evident in soft tissue strain injuries, which mainly manifest in the

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extramuscular and tendinous connective tissue. By contrast, isolated morphological damage to the muscle occurs only in a very few cases. Although the effectiveness of specific preventive and therapeutic approaches has yet to be fully established, the development and application of interventions targeting fascia offer interesting opportunities for coaches and therapists. Based on the available evidence, eccentric training and dynamic stretching exercises seem

to be capable of providing adequate stimuli. In DOMS, the connective tissue surrounding the muscle also seems to play an important role as the pain experienced by the individual is likely to originate from the fascia. While further research is still warranted definitively confirming the presence of fascial micro-ruptures, treatments aiming to ease the post-exercise phenomenon should be tailored according to the new findings.

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